Special Relativity Notes

Metric Tensors: General Definition, Properties and Spherical Coordinates

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1 Metric Tensors

Euclidean space (flat space)	**Minkowski space**
In 3D:	*In 3D:*
$d\vec{s} = dx\,\hat{e}_x + dy\,\hat{e}_y + dz\,\hat{e}_z$	$ds^2 = c^2dt^2 - d\vec{x}^{\,2}$
$ds^2 = dx^2 + dy^2 + dz^2$	
In n-D:	*In n-D:*
$ds^2 = \sum_{i=1}^{n} dx_i^2$	$ds^2 = g_{ij}\,dx^i dx^j$
	$= g_{11}\,dx^{1\,2} + g_{22}\,dx^{2\,2} + g_{33}\,dx^{3\,2}$

2 General Definition of the Metric Tensor

Q: How do we write down the metric in general?

$q^i \to$ contravariant coordinates, $\quad \vec{\epsilon}_i \to$ base vectors.

The displacement in Euclidean space per unit change in q^i is

$$\vec{\epsilon}_i = \frac{\partial x}{\partial q^i}\hat{e}_x + \frac{\partial y}{\partial q^i}\hat{e}_y + \frac{\partial z}{\partial q^i}\hat{e}_z$$

such that

$$\boxed{d\vec{s} = \vec{\epsilon}_i\,dq^i}$$

$$ds^2 = \sum_{i,j}(\vec{\epsilon}_i\,dq^i)\cdot(\vec{\epsilon}_j\,dq^j) = g_{ij}\,dq^i dq^j$$

where $\quad g_{ij} = \vec{\epsilon}_i \cdot \vec{\epsilon}_j \quad$ is called the **metric tensor** (in the covariant notation),

and, correspondingly,

$$g^{ij} = \vec{\epsilon}^{\,i} \cdot \vec{\epsilon}^{\,j} \qquad \text{(contravariant notation)}$$

3 Properties of the Metric Tensor

- $g_{ij} = g_{ji}$ (symmetric)
- Off-diagonal elements may not be zero.
- $g_{ik}\, g^{kj} = g_{ki}\, g^{ik} = \delta_i^j$
- g_{ij} and g^{ij} are **inverses** of each other.

So we can use the metric tensor to convert a contravariant vector to a covariant vector:

$$A^i = g^{ik} A_k, \qquad A_i = g_{ik} A^k$$

$$\vec{A} = \vec{\epsilon}_1 A^1 + \vec{\epsilon}_2 A^2 + \vec{\epsilon}_3 A^3 = \vec{\epsilon}^{\,1} A_1 + \vec{\epsilon}^{\,2} A_2 + \vec{\epsilon}^{\,3} A_3$$

The **dual (contravariant) basis vectors** are

$$\vec{\epsilon}^{\,i} = \frac{\partial q^i}{\partial x}\hat{e}_x + \frac{\partial q^i}{\partial y}\hat{e}_y + \frac{\partial q^i}{\partial z}\hat{e}_z$$

$$g_{ij} = \vec{\epsilon}_i \cdot \vec{\epsilon}_j, \qquad g^{ij} = \vec{\epsilon}^{\,i} \cdot \vec{\epsilon}^{\,j}$$

4 Orthogonality of the Dual Basis

$$\begin{aligned}\vec{\epsilon}^{\,i} \cdot \vec{\epsilon}_j &= \left(\frac{\partial q^i}{\partial x}\hat{e}_x + \frac{\partial q^i}{\partial y}\hat{e}_y + \frac{\partial q^i}{\partial z}\hat{e}_z\right) \cdot \left(\frac{\partial x}{\partial q^j}\hat{e}_x + \frac{\partial y}{\partial q^j}\hat{e}_y + \frac{\partial z}{\partial q^j}\hat{e}_z\right) \\ &= \frac{\partial q^i}{\partial x}\frac{\partial x}{\partial q^j} + \frac{\partial q^i}{\partial y}\frac{\partial y}{\partial q^j} + \frac{\partial q^i}{\partial z}\frac{\partial z}{\partial q^j}\end{aligned}$$

$$\vec{\epsilon}^{\,i} \cdot \vec{\epsilon}_j = \frac{\partial q^i}{\partial q^j} = \delta_j^i$$

5 Worked Example: Spherical Coordinates

Let $q^1 = r,\ q^2 = \theta,\ q^3 = \phi$:

$$x = r\sin\theta\cos\phi, \qquad y = r\sin\theta\sin\phi, \qquad z = r\cos\theta$$

Calculate $\vec{\epsilon}_r, \vec{\epsilon}_\theta, \vec{\epsilon}_\phi$.

5.1 Covariant basis vectors

$$\vec{\epsilon}_r = \frac{\partial x}{\partial r}\hat{e}_x + \frac{\partial y}{\partial r}\hat{e}_y + \frac{\partial z}{\partial r}\hat{e}_z$$

$$\vec{\epsilon}_r = \sin\theta\cos\phi\,\hat{e}_x + \sin\theta\sin\phi\,\hat{e}_y + \cos\theta\,\hat{e}_z$$

$$\vec{\epsilon}_\theta = \frac{\partial x}{\partial \theta}\hat{e}_x + \frac{\partial y}{\partial \theta}\hat{e}_y + \frac{\partial z}{\partial \theta}\hat{e}_z$$

$$\vec{\epsilon}_\theta = r\cos\theta\cos\phi\,\hat{e}_x + r\cos\theta\sin\phi\,\hat{e}_y - r\sin\theta\,\hat{e}_z$$

$$\vec{\epsilon}_\phi = -r\sin\theta\sin\phi\,\hat{e}_x + r\sin\theta\cos\phi\,\hat{e}_y$$

5.2 Contravariant (dual) basis vectors

Using $r = \sqrt{x^2+y^2+z^2}$, $\cos\theta = z/r$, $\tan\phi = y/x$:

$$\sin\theta\cos\phi = \frac{x}{r}$$

Calculate $\vec{\epsilon}^r, \vec{\epsilon}^\theta, \vec{\epsilon}^\phi$. Check g_{11}, g_{22}, g_{33}. Find ds^2.

$$\frac{\partial r}{\partial x} = \frac{1}{2}(x^2+y^2+z^2)^{-1/2}(2x) = \frac{x}{r}, \qquad \frac{\partial r}{\partial y} = \frac{y}{r}, \qquad \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\vec{\epsilon}^r = \sin\theta\cos\phi\,\hat{e}_x + \sin\theta\sin\phi\,\hat{e}_y + \cos\theta\,\hat{e}_z$$

(identical to $\vec{\epsilon}_r$, since $g_{rr} = 1$.)

For θ: from $\cos\theta = z/r$,

$$-\sin\theta\,\frac{\partial\theta}{\partial x} = \frac{\partial}{\partial x}\left(\frac{z}{r}\right) = -\frac{zx}{r^3} \quad\Longrightarrow\quad \frac{\partial\theta}{\partial x} = \frac{zx}{r^3\sin\theta}$$

$$-\sin\theta\,\frac{\partial\theta}{\partial y} = -\frac{zy}{r^3} \quad\Longrightarrow\quad \frac{\partial\theta}{\partial y} = \frac{zy}{r^3\sin\theta}$$

$$-\sin\theta\,\frac{\partial\theta}{\partial z} = \frac{1}{r} - \frac{z^2}{r^3} = \frac{r^2-z^2}{r^3} \quad\Longrightarrow\quad \frac{\partial\theta}{\partial z} = -\frac{\sin\theta}{r} \qquad \left(\because\ r^2 - z^2 = x^2+y^2 = r^2\sin^2\theta\right)$$

Substituting $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$:

$$\frac{zx}{r^3\sin\theta} = \frac{\cos\theta\cos\phi}{r}, \qquad \frac{zy}{r^3\sin\theta} = \frac{\cos\theta\sin\phi}{r}$$

$$\vec{\epsilon}^\theta = \frac{\cos\theta\cos\phi}{r}\hat{e}_x + \frac{\cos\theta\sin\phi}{r}\hat{e}_y - \frac{\sin\theta}{r}\hat{e}_z$$

For ϕ: from $\phi = \tan^{-1}(y/x)$,

$$\frac{\partial\phi}{\partial x} = \frac{1}{1+(y/x)^2}\left(-\frac{y}{x^2}\right) = -\frac{y}{x^2+y^2}, \qquad \frac{\partial\phi}{\partial y} = \frac{1}{1+(y/x)^2}\cdot\frac{1}{x} = \frac{x}{x^2+y^2}, \qquad \frac{\partial\phi}{\partial z} = 0$$

With $x^2+y^2 = r^2\sin^2\theta$:

$$-\frac{y}{x^2+y^2} = -\frac{\sin\phi}{r\sin\theta}, \qquad \frac{x}{x^2+y^2} = \frac{\cos\phi}{r\sin\theta}$$

$$\vec{\epsilon}^\phi = -\frac{\sin\phi}{r\sin\theta}\hat{e}_x + \frac{\cos\phi}{r\sin\theta}\hat{e}_y$$

5.3 Metric components and the line element

$$g_{ij} = \vec{\epsilon}_i \cdot \vec{\epsilon}_j$$

$$\begin{aligned} g_{11} = \vec{\epsilon}_r \cdot \vec{\epsilon}_r &= \sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta \\ &= \sin^2\theta + \cos^2\theta = 1 \end{aligned}$$

$$\begin{aligned} g_{22} = \vec{\epsilon}_\theta \cdot \vec{\epsilon}_\theta &= r^2\cos^2\theta\cos^2\phi + r^2\cos^2\theta\sin^2\phi + r^2\sin^2\theta \\ &= r^2(\cos^2\theta + \sin^2\theta) = r^2 \end{aligned}$$

$$\begin{aligned} g_{33} = \vec{\epsilon}_\phi \cdot \vec{\epsilon}_\phi &= r^2\sin^2\theta\sin^2\phi + r^2\sin^2\theta\cos^2\phi \\ &= r^2\sin^2\theta \end{aligned}$$

$$\boxed{ds^2 = dr^2 + r^2 d\theta^2 + r^2\sin^2\theta\, d\phi^2}$$